Rodrigo Andrade

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EDUCATION

University of Southern California, Viterbi School of Engineering, Los Angeles, CA **May 2027 (expected)**
Bachelor of Science in Electrical and Computer Engineering, GPA: 3.86
Presidential Scholar (Half-tuition award based on academic performance and extracurriculars)
Honors Thesis, Robot Learning **Fall 2026 – May 2027 (expected)**
Professional Director, Brazilian Student Association (BRASA) USC **August 2023 – May 2025**

- Selected to serve on the BRASA USC executive board, supporting Brazilian students' professional development.
- Organized recruiting and networking events with American and Brazilian companies.

WORK EXPERIENCE

Physical Superintelligence (PSI) Lab, Undergraduate Researcher, Los Angeles, CA **August 2025 – Present**

- Built and deployed teleoperation and multimodal data collection systems (RealSense cameras, synchronized state/action logging) for real-robot experiments, enabling high-quality human demonstrations for manipulation tasks.
- Trained and evaluated imitation learning and diffusion-based policies for robotic manipulation; set up and integrated robot arms from scratch to support bimanual manipulation experiments.
- Currently developing a Dex Umi-style dexterous data collection hand, including mechanical assembly and adaptation of existing designs, and continuing robot learning research this semester.

Dynamics Robotics and Control Laboratory, Undergraduate Researcher, Los Angeles, CA **October 2024 – Present**

- Contributed to the mechanical and electromechanical design of a full-scale humanoid robot, including shoulder and knee motor interfaces and foot design.
- Implemented an inverse kinematics solver for the humanoid's arms within a modular ROS 2 control architecture, enabling coordinate-based motion commands.
- Supported hardware experiments evaluating reinforcement-learning-based walking policies, including robot bring-up, testing, and debugging.
- Helped design Hector Mini, a fully 3D-printed bipedal humanoid robot featuring a novel rolling-contact knee joint.

Rio de Janeiro City Education Department, Teacher, Rio de Janeiro, RJ **June – July 2024**

- Selected to teach robotics to over 100 elementary school students (ages 8-11) as part of Rio de Janeiro's new STEM curriculum initiative.
- Developed age-appropriate instructional material (basic CAD, electronics, soldering, circuits, and programming) to introduce engineering thinking and problem-solving in resource-constrained environments, and conducted technical training sessions for other teachers to support curriculum rollout and long-term program sustainability.

SIATT (High-Technological Integrated Systems), Intern, São José dos Campos, SP, Brazil **May – June 2024**

- Worked in the simulation and control team on the MANSUP naval missile program, a waypoint- and radar-guided system, contributing to physics-based simulation environments for missile dynamics and trajectory evaluation.
- Assisted in control algorithm validation and testing through simulation-based analysis.

Grupo Q (Qconcursos), Intern, Rio de Janeiro, RJ **May – July 2023**

- Designed and deployed an intelligent study schedule generator leveraging Google APIs and OpenAI APIs.
- Built the full production system using Cloudflare Workers, focusing on scalability and low-latency deployment.
- Translated user requirements into an end-to-end software architecture, from API integration to deployment.

Buntech, Apprentice, Indaiatuba, SP **March 2023**

- Rotated through multiple engineering roles in a manufacturing and industrial environment, including lab operations, project engineering, maintenance, and industrial engineering.
- Assisted with testing, troubleshooting, and process support across different stages of production.

ACTIVITIES AND LEADERSHIP

Projeto++, Co-Founder, Rio de Janeiro, RJ **November 2021 – Present**

- Co-founded and led a nonprofit educational robotics initiative providing free robotics education to public school students in Brazil.
- Secured R$10,000 in funding and led a fundraising effort resulting in computer donations valued at approximately USD $20,000.
- Designed and delivered weekly hands-on robotics lectures, reaching 100+ students to date.
- Established the project's online presence at www.projetomaismais.com, increasing visibility and long-term engagement.
- Ensured organizational sustainability by training and transitioning leadership to 15+ high school students, now forming the project's team in Brazil.

The British School Robotics Club, Co-Founder, Rio de Janeiro, RJ **June 2022 – December 2022**

- Co-founded the school's first robotics club, creating a formal space for students to develop technical skills in robotics and engineering.
- Raised R$6,000 from parents and school supporters to fund equipment and activities.
- Established the organizational structure and curriculum before graduation, enabling long-term continuity; the club remains active today with 20+ participating students.

PROJECTS

SKILLS

- **Languages:** Portuguese (native), English (fluent), Spanish (conversational)
- **Skills:** C++, Python, ROS 2, MuJoCo, Isaac Sim, PyTorch, embedded systems programming, soldering, 3D printing
- **Applications:** CAD (Fusion 360 and SolidWorks), PCB design, MATLAB

AWARDS

- Engineering Mathematics Prize awarded for outstanding performance in Calculus (2/400 awarded) [2024]
- Dean's List, Viterbi School of Engineering, USC (4 times)
- Gold Medal, Brazilian Astronomy Olympiad [2021, 2020, 2019]; Gold Medal, Brazilian Science Olympiad [2021]; Bronze Medal, Brazilian Science Olympiad [2020]; Honorable Mention, Brazilian Mathematical Olympiad of Public Schools (OBMEP) [2019]